

	ESTIMASI KALKULASI						
	Parameter	Symbol	Formula	Calculation	Estimated Value	Unit	Description
	Vehicle Mass	m_vehicle	Assumption	-	85	kg	Estimated vehicle weight without driver
	Driver Mass	m_driver	SEM Regulation	-	70	kg	Minimum driver weight according to Shell Eco-marathon regulations
	Total Vehicle Mass	m	$m = m_{\text{vehicle}} + m_{\text{driver}}$	$85 + 70$	155	kg	Total mass used in vehicle dynamics calculations
	Air Density	ρ	Standard atmosphere	-	1.225	kg/m ³	Air density at sea level
	Drag Coefficient	Cd	Assumption	-	0.1	-	Estimated aerodynamic drag coefficient of a competitive SEM Urban Concept vehicle
	Vehicle Width	W	Assumption	-	1.22	m	Estimated within SEM regulations
	Vehicle Height	H	Assumption	-	1.05	m	Estimated within SEM regulations
	Shape Factor	k	Assumption	-	0.6	-	Correction factor for streamlined body shape
	Frontal Area	A	$A = W \times H \times k$	$1.22 \times 1.05 \times 0.60$	0.77	m ²	Estimated projected frontal area
	Rolling Resistance Coefficient	Crr	Assumption	-	0.004	-	Typical value for low rolling resistance SEM tyres
	Wheel Diameter	D	Assumption	-	16	inch	Selected within SEM regulations (14–18 inch)
	Wheel Radius	r	$r = (D \times 0.0254) / 2$	$(16 \times 0.0254) / 2$	0.203	m	Effective wheel radius

	Cruise Speed	v	Assumption	-	30	km/h	Target average cruising speed
	Cruise Speed (SI)	v	$v = 30 / 3.6$	$30 / 3.6$	8.33	m/s	Converted to SI unit
	Aerodynamic Drag Force	Fd	$F_d = 0.5 \times \rho \times C_d \times A \times v^2$	$0.5 \times 1.225 \times 0.10 \times 0.77 \times 8.33^2$	3.27	N	Aerodynamic drag force
	Rolling Resistance Force	Fr	$F_r = C_{rr} \times m \times g$	$0.004 \times 155 \times 9.81$	6.08	N	Tyre rolling resistance force
	Total Resistance Force	Fres	$F_{res} = F_d + F_r$	$3.27 + 6.08$	9.35	N	Total resistance during cruising
	Cruise Mechanical Power	Pmech	$P = F \times v$	9.35×8.33	77.9	W	Mechanical power required at the wheels
	Overall Drivetrain Efficiency	η	Assumption	-	92	%	Combined efficiency of motor, controller and drivetrain
	Cruise Electrical Power	Pelec	$P_{elec} = P_{mech} / \eta$	$77.9 / 0.92$	84.7	W	Electrical power supplied to the drivetrain
	Target Acceleration	a	$a = \Delta v / t$	$8.33 / 10$	0.833	m/s ²	Assumed acceleration from 0–30 km/h in 10 s
	Acceleration Force	Facc	$F = m \times a$	155×0.833	129.1	N	Force required for vehicle acceleration
	Total Force During Acceleration	Ftotal	$F_{total} = F_{acc} + F_d + F_r$	$129.1 + 3.27 + 6.08$	138.45	N	Total tractive force required during acceleration
	Required Wheel Torque	T	$T = F \times r$	138.45×0.203	28.1	Nm	Wheel torque required during acceleration
	Peak Mechanical Power	Ppeak	$P = F \times v$	138.45×8.33	1.15	kW	Estimated peak mechanical power during acceleration